

Hey, everybody. Welcome to your chapter 18 review, another in our series of integrated training components. This time, we'll focus on plyometric training, also called jump or reactive training. Although plyometrics are not usually stressed in the implementation of a fitness program for personal training clients, it's a critical concept in human movement that can be included into the overall training plan. And the more you think about its benefits, the more important it becomes for more than just athletes but all active individuals, considering the many injuries that occur as a result of a person's inability to control decelerative forces imparted on the body.

Plyometric training develops efficient control and production of ground reaction forces, which are used to project the body with greater velocity or speed of movement. The ultimate goal is to maximize function, which also results in increased speed of movement. Clients must possess adequate core strength, joint stability, range of motion, and balance to perform explosive plyometric exercises. It would not be an appropriate form of training for individuals with select chronic diseases or other functional limitations.

When you finish this chapter, you should be able to explain plyometric training and its three phases, summarize the benefits of plyometric training for different client types with differing health goals, identify proper progressions of plyometric training exercises, and employ methods used to effectively execute, instruct, and cue plyometric training exercises. And now on to the good stuff.

So first of all, what is plyometric training? Plyometric training is a form of exercise that uses explosive movements such as bounding, jumping, or powerful upper body movements to develop muscular power. The ability to generate force as quickly as possible is known as rate of force production. Success in everyday activities and sports depends on the speed at which muscular force is generated.

Plyometric exercise uses a characteristic of muscles known as the stretch-shortening cycle. This cycle improves the elastic properties of connective tissue and muscle fibers by enhancing stored elastic energy through the eccentric or lengthening phase and releasing it during the concentric or shortening phase to increase force production. In fact, the integrated performance paradigm states that to move with precision, forces must be eccentrically loaded, isometrically stabilized, and then unloaded or concentrically accelerated.

There are three distinct phases of the stretch-shortening cycle involved in a plyometric exercise, including the eccentric or loading phase, the amortization or transition phase, and the concentric or unloading phase. During the eccentric phase, elastic energy is stored and muscle spindles are stimulated, which sends signals to the central nervous system. It's a preloading phase, like stretching a rubber band, where potential energy is stored in the elastic components of the muscle and is predicated on the magnitude of stretch, the rate of stretch, and the duration of the stretch. Any manipulation of these variables can enhance the amount of stored energy.

The amortization phase is the transition period or electromechanical delay that describes the space and time during which the muscle is switching from overcoming force eccentrically to imparting force in the intended direction. The shorter this amortization phase, the more effective and powerful the plyometric movement, as the stored elastic energy is used efficiently in the transition.

The concentric phase is the body's response to what occurs during the preceding eccentric and amortization phases, including the stored elastic energy, to either enhance muscle-force production and deliver an explosive action or dissipate energy as heat. These types of explosive muscular contractions and all three stages can be seen in practical instances like rebounding in basketball.

Watch trained basketball players as they jump up for a loose ball, and you'll see them prepare by lowering their body slightly and flexing at the ankles, knees, and hips. This is the eccentric loading phase. You may even see players drop their arms to assist in takeoff. At a fairly shallow point, players will reverse this downward motion and rapidly project themselves up from the ground, extending their ankles, knees, hips, and arms upward. This is the concentric phase. The overall height of their jump is determined by their vertical velocity or how fast they leave the ground.

By integrating plyometric training, clients will be able to generate greater power output responses at faster speeds. Scientific literature indicates enhanced muscle power has been associated with improved athletic sports performance through improved jump performance, reduced sprint times, and increased strength. Plus, the use of plyometric training prepares athletes and clients for the acceleration, deceleration, and change-of-direction demands in certain sports-- and activities of daily living, for that matter. Chasing after a couple of toddlers can require the same athletic ability as those basketball players I just mentioned.

Plyometric training exercises must be implemented in a systematic and progressive fashion to ensure safety and minimize the chance of injury. A thorough assessment may help distinguish clients who may or may not be prepared for this type of training. A client must exhibit basic levels of total body strength, core strength, and balance to participate, and wear supportive footwear when they do.

The training service should also be selected carefully, as it can have a profound effect on performance. Exercises should progress from simple to moderate to advanced movements and from low intensity to medium intensity to high intensity. Frequency and recovery time should also be carefully considered. A general recommendation is to allow at least one day between intense plyometric sessions. However, for novice individuals, that may need to be extended to at least 48 to 72 hours between sessions. Between drills, 60 to 120 seconds should be allotted to provide sufficient time for full recovery.

Mastering and demonstrating proper techniques is important for every move you teach, particularly during jumping and landing. You'll base your client's plyometric program intensity and volume on their ability to execute the exercises and maintain adequate training technique. If technique is lost, the intensity and possibly volume should drop until proper technique is achieved.

When you introduce plyometric exercises, especially to new or beginner clients, the movements should initially involve small jumps to best learn the movement pattern. They are intended to establish optimal landing mechanics, postural alignment, and eccentric strength, also known as the ability to decelerate. When an individual lands during these exercises, they should hold the landing position for three to five seconds to promote stabilization and reinforce proper alignment. During this time, make any adjustments necessary to correct faulty postures and lower body alignment before they perform the next jump. An example of this type of exercise would be a squat jump with stabilization.

The next progression should involve jumps with more amplitude and dynamic motion. The speed can also be increased. These exercises are intended to improve dynamic joint stabilization, eccentric strength, rate of force production, and coordination of the entire human body. Contrary to the progressions just discussed, where individuals hold the landing position between jumps, you'll have the individual perform these exercises in a repetitive manner, spending a relatively short amount of time on the ground before repeating the drill. Examples for this progression would be a squat jump or tuck jump.

And finally, the last progression introduces exercises that involve explosive, powerful movements. These exercises are intended to further improve rate of force production, eccentric strength, and reactive joint stabilization, and are to be performed as fast and explosively as possible. Some examples within this progression are ice skaters, single-leg power step-ups, and depth jumps.

In summary, though plyometrics are often associated with a more athletic population, if you understand the art and science of progression and regression of exercise, then plyometric training can be scaled and modified to suit many clients' capabilities and goals. Just make sure you are communicating effectively so your clients not only understand where plyometrics fit in relation to their goals but also the benefits and importance of knowing and performing proper landing and rebounding mechanics.

For example, if a client doesn't understand proper foot contact, form, and alignment, then they won't be able to execute it either. And when they're ready for greater challenge and demands, you can always add more dynamic motions and change the tempo and speed, effectively building off the solid foundation you've established them. With that, I will leave you to your own dynamic training of the mind as you proceed to the next chapter. Until next time, be well, and happy studying.